

Problem 2

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Problem 1 Determine the following indefinite integrals.

(a)

$$\int (\sec^2(x) + 5) dx$$

(b)

$$\int (\sec^2(x) - \sec(x) \tan(x)) dx$$

(c)

$$\int \frac{x + 3x^5}{x^3} dx$$

(d)

$$\int \frac{1 + 2x}{1 + x^2} dx$$

(HINT: Think about the derivative of $\ln(f(x))$).

(e)

$$\int \frac{2 + x^2}{1 + x^2} dx$$